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Patterns and drivers of global groundwater drought recovery

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Abstract

Groundwater resources are under increasing pressures, due to changes in climatic and anthropogenic drivers. Meanwhile extreme drought events can strongly affect the resilience of groundwater systems, due to decreased groundwater recharge and increased groundwater abstraction to meet increasing water demands. As groundwater systems tend to respond on a longer timescale, it is important to understand how these systems recover from these drought events back to normal levels. This study provides the first global study of groundwater drought recovery based on high-resolution simulations of a global groundwater model. We find that the spatial variability of the groundwater drought recovery rate is mostly influenced by the local climate, while local geophysical attributes such as topography and aquifer storage capacities can account for regional differences. We introduce a classification of groundwater drought recovery regimes (resilient, stable, vulnerable, and unstable) based on drought event-specific recovery rates, which allows us to identify the resilience of groundwater systems around the world. We observe that areas sharing similar average recovery rates can differ in their underlying recovery regime and controlling drivers. Resilient regions, covering the majority of locations (57%), exhibit rapid and robust recovery following droughts. Vulnerable (15%) and stable locations are shaped by geophysical constraints and heightened climate variability, whereas unstable regions (26%) are characterised by frequent, successive drought events leading to a reduced drought recovery capacity. By systematically relating recovery regimes to drought characteristics, environmental and anthropogenic factors, our findings offer insights into the mechanisms that influence groundwater drought resilience. These results show that integrating regional climatic, geophysical and anthropogenic aspects into groundwater management practices is important and could potentially support the identification of regions at risk of crossing critical thresholds and support the development of targeted strategies to enhance global groundwater resilience in the face of climate and anthropogenic changes.

1. Introduction

Groundwater is a critical source of freshwater and a vital component of the global water cycle, supporting ecosystems and fulfilling the water demands for agriculture, industry, and domestic water use, accounting for approximately 35% of all freshwater withdrawals (Döll *et al* 2012, Condon *et al* 2021, Scanlon *et al* 2023). In many regions, it serves as a primary source of drinking water and sustains agricultural production, particularly where surface water is unavailable or intermittent (Famiglietti and Rodell 2013, Leijnse *et al* 2024). Beyond its role in regular supply, groundwater is especially important as a source of freshwater during droughts, acting as a natural buffer, when other sources diminish quickly or in areas with high water stress (Wada *et al* 2010). With the rising demand and pressure on these resources, groundwater systems have seen an acceleration in groundwater level declines, driven by both anthropogenic and climatic drivers (Famiglietti 2014, Wanders *et al* 2015, Wu *et al* 2020b, Jasechko *et al* 2024). In addition to this decline in groundwater levels, we observe strong trends in drought frequency which puts extra pressure on these vital resources (Wanders and Wada 2015). Therefore, it is

important to understand how the resilience of groundwater resources is affected by extreme events such as droughts, and how these systems recover post-drought, whether they are more or less resilient, and whether they might potentially lose their buffer function in the future.

Droughts typically propagate from meteorological drought (reduced precipitation) to agricultural drought (declining soil moisture) to hydrological drought (affecting streamflow and groundwater, Mishra and Singh 2011, Van Loon and Laaha 2015). The timescales over which this propagation occurs vary considerably across system types and are affected by both the local climatology and geophysical properties (Van Lanen *et al* 2013). Especially for groundwater, these timescales can be of importance, as groundwater systems tend to respond more slowly than surface water, and therefore storage deficits can persist long after the triggering precipitation deficit has ended, creating a drought memory that shapes the starting conditions for drought recovery (Bloomfield and Marchant 2013, Cuthbert *et al* 2019). However, slower responses can also provide a buffer against drought impacts, with some studies showing groundwater's resilience to drought (MacDonald *et al* 2019, MacAllister *et al* 2020). Existing studies highlight that recovery from groundwater drought depends on a range of factors, such as aquifer properties, recharge conditions, and pre- and post-drought climate forcing, as well as local geophysical characteristics, with recovery times ranging from months to decades depending on locations (Hellwig *et al* 2020, Wu *et al* 2020a, Alam *et al* 2021, Schreiner-McGraw and Ajami 2021, Chen *et al* 2024). Furthermore, previous studies have focused on how geophysical properties such as climate, geology and aquifer properties, land use and elevation, as well as drought severity, intensity and duration of the meteorological drought can play a role in drought propagation to groundwater systems (Wong *et al* 2012, Van Lanen *et al* 2013, Van Loon 2015, Schreiner-McGraw and Ajami 2021). However, it remains largely unknown how these factors affect the recovery of groundwater droughts, especially for groundwater at the global scale, as observations are limited and models often provide information at spatial resolutions that are too coarse. We hypothesize that the same factors that govern how droughts propagate into groundwater systems also influence how, and how quickly, those systems recover once precipitation returns to normal. Ultimately, making an understanding of propagation dynamics a prerequisite for assessing recovery.

Typical drought recovery follows a pattern with precipitation returning to normal, followed by a recovery in soil moisture, vegetation, and surface water. However, groundwater resources usually lag behind these other components, due to slower recharge processes and aquifer properties (Bloomfield and Marchant 2013, Cuthbert *et al* 2019). The hydrological conditions prior to a drought event also impact the recovery potential. If a groundwater system has not fully recovered from a preceding drought, it enters the next event in an already depleted state, decreasing its resilience and potentially accelerating the development of subsequent droughts (Van Loon *et al* 2014, Bloomfield *et al* 2019). In addition, increased human influence affects the propagation and recovery processes by intensifying water shortages from groundwater abstraction, on top of the natural precipitation deficit stemming from the meteorological drought (Mishra and Singh 2011, Akl *et al* 2025). Especially, if groundwater abstractions are unsustainable over the long term, meaning higher than the natural system's capacity to recover via recharge, they can affect the resilience of groundwater systems, leading to increased water scarcity issues and intensified drought impacts (Famiglietti *et al* 2011, Castle *et al* 2014, Famiglietti 2014, Bierkens *et al* 2021, Akl *et al* 2025). This is particularly relevant in heavily irrigated areas, where groundwater is often used as the primary fallback when surface water availability declines during drought, creating a feedback whereby drought conditions drive the very abstraction that impacts recovery (Wada 2016, Ren *et al* 2024). Changes in groundwater storage, which might likely affect the recovery process as well, for regions with high groundwater abstractions can be related to both climate variability as well as human influence, which is ultimately often directly influenced by changes in precipitation and drought (Asoka *et al* 2017). With groundwater droughts also being affected by both natural fluctuation, due to climate and seasonality (Mishra and Singh 2011, Van Loon *et al* 2014, Bloomfield *et al* 2019, Akl *et al* 2025), and anthropogenic drivers (Famiglietti 2014), it is therefore essential to not only understand groundwater drought development but also better understand the drought recovery processes.

Despite a lot of research on drought propagation being done on various scales (Peters *et al* 2006, Van Loon *et al* 2012, Hellwig *et al* 2020, Teutschbein *et al* 2025), the understanding of groundwater drought recovery remains limited, particularly at regional to global scales (Alam *et al* 2021, Schreiner-McGraw and Ajami 2021). Existing studies show that recovery from groundwater drought depends on a range of factors, such as aquifer properties, recharge conditions, and pre- and post-drought climate forcing, as well as local geophysical characteristics, with recovery times ranging from months to decades depending on locations (Hellwig *et al* 2020, Wu *et al* 2020a, Alam *et al* 2021, Schreiner-McGraw and Ajami 2021, Chen *et al* 2024). While high precipitation during post-drought periods can accelerate recovery, continued groundwater abstraction often constrains both the degree and rate of recovery, especially in heavily irrigated regions (Alam *et al* 2021). The water storage component as well as water management strategies

therefore also play an important role in groundwater drought recovery, and are often studied more at a local scale (Parry *et al* 2016b, Wendt *et al* 2020, Alam *et al* 2021).

While local studies provide valuable insights, groundwater systems often span large regions and cross political boundaries, necessitating large-scale assessments either by modelling efforts or large-scale observation assessments. What is currently missing is a spatially consistent, global-scale framework that can characterise the relative rates of drought development and recovery, and classify locations based on their drought resilience. This information can help identify regions at risk and guide water management under future pressures. For such a framework, spatially continuous, long-term groundwater data provided by global modelling approaches are an ideal basis. Data limitations exist, both in terms of limited local groundwater observations and consistent modelling frameworks that can quantify drought dynamics. Such a framework is currently hampered by uncertainties in model parametrisation and computational challenges. Alternatively, large-scale observational datasets such as the Gravity Recovery and Climate Experiment (GRACE) can be used to derive terrestrial water storage anomalies, that provide insights into groundwater variability and depletion (Döll *et al* 2014, Thomas *et al* 2014). However, their coarse spatial resolution, sensitivity and variability to processing uncertainties may limit their ability to fully assess groundwater drought recovery processes (Scanlon *et al* 2023, Akl *et al* 2025).

Physically-based global groundwater models can overcome some of these limitations by providing spatially continuous, process-based estimates of groundwater storage and fluxes at higher resolution, and by explicitly representing human influences such as groundwater abstraction, offering a complementary approach (Bierkens *et al* 2015, Clark *et al* 2015, Condon *et al* 2021, Gleeson *et al* 2021). Recent advances in these modelling techniques include incorporating groundwater processes into global land surface and earth system models, with large-scale analyses covering continental and global scales, and studies evaluating and demonstrating the potential of models to simulate groundwater levels, pumping limits, and interactions with river systems (Maxwell *et al* 2015, de Graaf *et al* 2019, 2024, Bierkens *et al* 2021). In recent years, these models have overcome significant computational limitations and been updated and improved to simulate groundwater resources at hyper-resolution (1 km) and allow for assessing groundwater resources under different climate and socio-economic scenarios (Verkaik *et al* 2024, van Jaarsveld *et al* 2026).

The new advances in groundwater modelling now allow us, for the first time, to study the development and recovery of groundwater droughts at a spatial scale relevant to groundwater dynamics. This would provide a comprehensive understanding of the large-scale patterns and underlying processes governing groundwater drought recovery. Understanding these patterns and processes can help better estimate future groundwater developments, especially in regions with sparse observational datasets. The use of large-scale groundwater models offers the chance to systematically identify, compare, and classify recovery behaviours across diverse hydroclimatic and anthropogenic settings. The objective of this study is to characterize and quantify the spatial patterns of groundwater drought recovery at the global scale. We aim to understand the spatial patterns and underlying drivers, and classify the observed behaviours into recovery regimes to better comprehend future groundwater drought recovery processes and trends, and connect locations with similar recovery dynamics. To address these aims, we defined and analysed drought events based on groundwater simulations from the global groundwater model GLOBGM (v1.1) at 1 km resolution (van Jaarsveld *et al* 2026). We identified drought events and their characteristics (e.g. including drought duration, intensity, pre- and post-event dynamics) and calculated the drought recovery rates. To understand the recovery dynamics and potential drivers, we introduced a recovery regime classification that follows ideas of system resilience introduced by Hashimoto *et al* (1982) for water resources systems, where resilience describes how quickly a system recovers from failure or impact. We applied these concepts specifically to the groundwater system by including drought event-specific rates that describe pre-event, development, recovery, and post-event conditions. Based on these insights, we classified locations into resilient (slow development, fast recovery), stable (equal development and recovery), vulnerable (fast development, slow recovery) and unstable (re-entering drought conditions) recovery regimes. Combining the latter with geophysical, climatological and anthropogenic aspects by incorporating model-specific information additionally highlighted underlying processes and drivers.

2. Material and methods

The following sections contain an overview of the global groundwater simulations used (section 2.1), the drought recovery definition and drought characteristics (section 2.2), as well as the approach of the recovery regime classification (section 2.3).

2.1. Global groundwater simulations

The groundwater drought recovery analysis is based on monthly water table depth simulations from the high-resolution global groundwater model GLOBGM (v1.1) by van Jaarsveld *et al* (2026). The model is based on a global two-layer transient MODFLOW-based model at 30 arc-seconds (1 km) spatial resolution. The two-layer configuration represents unconfined, confining, or confined aquifer systems (Verkaik *et al* 2024). For cells where a confining layer is present, the top model layer represents the confining layer, and the bottom layer is the confined aquifer; in regions where a confining layer is absent, the bottom model layer represents the unconfined aquifer. GLOBGM (v1.1) is coupled to the land-surface component of PCR-GLOBWB2.0, a global hydrological model at 5 arc-minutes spatial resolution (Sutanudjaja *et al* 2018). With the coupling to PCR-GLOBWB2.0, GLOBGM (v1.1) includes fluxes such as local runoff, groundwater recharge, and groundwater abstraction. The latter is especially interesting to understand and estimate the anthropogenic influence on groundwater systems. The GLOBGM (v1.1) is the latest, updated global groundwater model, based of the original global groundwater model by de Graaf *et al* (2015, 2017) and the follow-up, high-resolution version GLOBGM (v1.0) by Verkaik *et al* (2024). With the newest version, GLOBGM (v1.1) includes model calibration and validation, as well as improved drainage routines, downscaling of input variables such as groundwater recharge from PCR-GLOBWB2.0, addressing previously listed sources of uncertainties in large-scale groundwater models (Reinecke *et al* 2025). With all those updates, it currently provides one of the few dynamic global groundwater simulations at spatial resolutions below 10 km. Validation against observation wells, using the $KGE - NP_{\text{skill}}$ score, showed that the model is able to simulate groundwater levels in shallow (0–20 m) and intermediate depth (20–60 m) with 80% of observation wells considered skilful, and a slight decrease to 70% for depths deeper than 60 m (van Jaarsveld *et al* 2026). Furthermore, trend analysis highlighted that the model is able to simulate groundwater depletion as well as increases for known cases (Jasechko *et al* 2024). For more detailed information on the updated, high-resolution groundwater model GLOBGM (v1.1), we refer to van Jaarsveld *et al* (2026). For this study, the monthly water table depth simulations (including groundwater abstractions) from 1960–2019 are used for the groundwater drought recovery analysis. These correspond to the historical reference simulations provided by van Jaarsveld *et al* (2026), a reanalysis-based reconstruction of groundwater based on GSWP3-W5E5 forcing input Lange and Büchner (2021).

2.2. Drought analysis: recovery definition and drought characteristics

For the groundwater drought recovery analysis, we follow the hydrological drought termination definition by Margariti *et al* (2019), based on Parry *et al* (2016a). The concept introduced for streamflow uses a threshold approach to identify drought events (start, end and total drought event duration). In this study, we adopt that methodology but apply it to the groundwater timeseries.

For every simulated grid cell from the GLOBGM (v1.1) simulations, we compute the mean and standard deviation of the groundwater table depth for every month over the whole time period 1960–2019. The threshold is defined as one standard deviation above the mean, which results in a monthly variable threshold. According to the Akaike Information Criterion used, we checked and found that in 54% of the world a normal distribution fits monthly groundwater levels better than gamma or log-normal distributions. This means that the assumption of normality for monthly groundwater levels is preferable to the available alternatives. Furthermore, due to the size and computational demands of the hyper-resolution data, using an empirical distribution was not an option, as the computational cost would be too high. Therefore, we opted to use a normal distribution with a mean and standard deviation for our threshold globally. This also ensures we have spatial consistency in drought patterns and no artefacts in the drought recovery patterns, which could arise from differences in drought quantification method and specifically the distribution used.

A period is classified as a drought event (figure 1) when the groundwater table depth is higher than the threshold value of one standard deviation. Using one standard deviation results in a threshold of about 16% of the highest water table depths that are taken into account as drought events. This is in line with, for example, the drought threshold used by the European Drought Observatory and will result in a balance between the number of events and the severity of the selected events.

For each drought event (t_{start} to t_{end}), the moment of the largest difference in groundwater table depth relative to the threshold was identified (t_{maxInt} at Δ_{maxInt}). The total drought duration (t_{tot}) in months (*month*) is given by:

$$t_{\text{tot}} = t_{\text{end}} - t_{\text{start}} \quad (1)$$

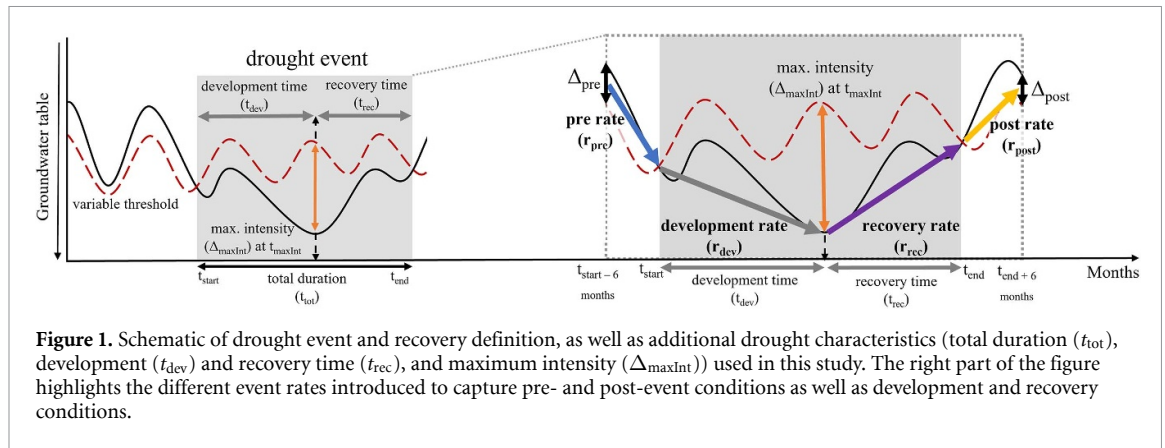


Figure 1. Schematic of drought event and recovery definition, as well as additional drought characteristics (total duration (t_{tot}), development (t_{dev}) and recovery time (t_{rec}), and maximum intensity (Δ_{maxInt})) used in this study. The right part of the figure highlights the different event rates introduced to capture pre- and post-event conditions as well as development and recovery conditions.

with the start of the event (t_{start}) and end of the drought event (t_{end}). Similarly, the drought development time in months (*month*) can be obtained with:

$$t_{dev} = t_{maxInt} - t_{start} \quad (2)$$

and the drought recovery time (t_{rec}) in months (*month*) is then given by:

$$t_{rec} = t_{end} - t_{maxInt} \quad (3)$$

which leads to a drought recovery rate (m month^{-1}):

$$r_{rec} = \frac{\Delta_{maxInt} - \Delta_{end}}{t_{rec}} \quad (4)$$

where Δ_{maxInt} and Δ_{end} are the difference between the threshold and the groundwater table at t_{maxInt} and t_{end} in meters (*m*) respectively.

We also tracked pre- and post-event conditions by calculating pre- and post-rates over a six-month window before and after a drought event. The pre-event rate (m month^{-1}) is given by:

$$r_{pre} = \frac{\Delta_{pre} - \Delta_{start}}{6} \quad (5)$$

where Δ_{pre} is the difference between the threshold and groundwater table at $t_{start} - 6$ and Δ_{start} is the difference between the threshold and groundwater table at t_{start} . The post-event rate (m month^{-1}) is given by:

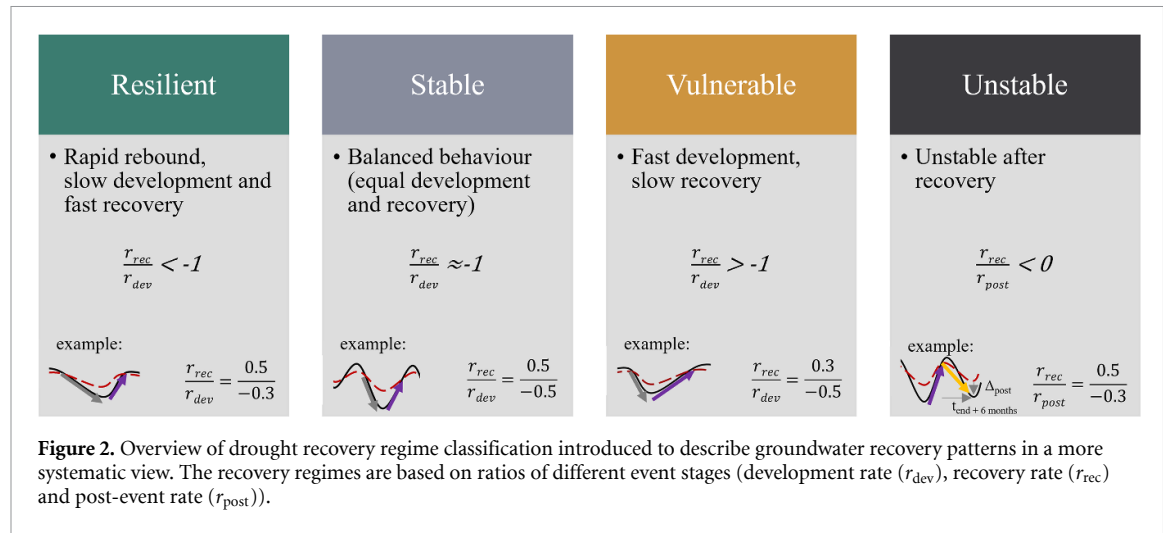
$$r_{post} = \frac{\Delta_{end} - \Delta_{post}}{6} \quad (6)$$

where Δ_{post} is the difference between the threshold and groundwater table at $t_{end} + 6$ and Δ_{end} is the difference between the threshold and groundwater table at t_{end} . Finally, the development rate (m month^{-1}) was calculated as:

$$r_{dev} = \frac{\Delta_{start} - \Delta_{maxInt}}{t_{dev}} \quad (7)$$

which provides the rate of drought development and can be compared with r_{rec} to assess whether there is symmetry in drought development and recovery. All these event rates combined describe how quickly a drought event approaches (r_{pre}), reaches its maximum intensity (r_{dev}), recovers (r_{rec}), and whether it fully recovers or transitions into a next event (r_{post}).

To connect the groundwater recovery rate (r_{rec}) to other drought characteristics, we also computed the number of events and the duration between events (event gap, t_{gap}), in addition to the previously described event rates (r_{pre} , r_{dev} , r_{rec} , r_{post}), maximum intensity (Δ_{maxInt}), and drought times (total duration (t_{tot}), development time (t_{dev}), and recovery time (t_{rec})). Before calculating the average of all these drought characteristics to be able to look at the spatial pattern for every location, we filtered out locations where drought events did not meet certain criteria. Droughts that occurred at the start or end of the simulation period were excluded to avoid affecting the statistical summary. These events and locations were often linked to areas with strong groundwater trends, as highlighted by



(van Jaarsveld *et al* 2026). While locations with weaker trends remain, the main selection might under-represent places where groundwater abstraction might dominate drought vulnerability. Second, we dropped shorter droughts of $\leq$ two months, as recovery rates (r_{rec}) for such events would not be reliable, as it would be hard to capture these very short droughts with a monthly time step. The average drought characteristics are used to perform a global analysis and to relate different drought characteristics to groundwater drought recovery rates (r_{rec}). To compare locations with different climatologies and assess the effect of year-to-year variations in climatic drivers, we standardized the recovery rate (r_{rec}) (and other event rates) by the standard deviation of the groundwater table depth as a measure of the long-term dynamic range (supplementary figure 2). This helps to focus on the underlying recovery pattern that remains after removing the location-specific internal hydrologic variability.

By first identifying drought events from the groundwater simulations from GLOBGM (v1.1) using a variable threshold approach, we were able to characterise every drought event based on their duration (t_{tot}), maximum intensity (Δ_{maxInt}), as well as their development and recovery times (t_{dev} and t_{rec}). Based on that, we could extend the drought characterisation including drought event rates at different stages throughout the drought (pre-event rate (r_{pre}), development rate (r_{dev}), recovery rate (r_{rec}), and post-event rate (r_{post})). These event rates will serve as the building blocks for the recovery regime classification.

2.3. Drought analysis: recovery regime classification

To understand and connect the spatial drought recovery pattern to potential underlying drivers and drought dynamics, we introduce a classification approach that describes drought recovery into different recovery regimes based on average event behaviour (event rates) (figure 2).

We use the ratios of the different event rates (pre-event (r_{pre}), development (r_{dev}), and post-event (r_{post})) to describe how locations behave relative to the recovery rate (r_{rec}).

For the development ratio, defined as:

$$R_{dev} = \frac{r_{rec}}{r_{dev}} \quad (8)$$

a value of $R_{dev} = -1$ indicates that the development rate (r_{dev}) equals the recovery rate (r_{rec}). A value $\ll -1$ means that recovery is significantly faster than development, while a value $\gg -1$ indicates that the development is faster than the recovery. The same pre-ratio and post-ratio are given by:

$$R_{pre} = \frac{r_{rec}}{r_{pre}} \quad (9)$$

$$R_{post} = \frac{r_{rec}}{r_{post}} \quad (10)$$

where $R_{pre} = -1$ and $R_{post} = 1$ indicating balanced pre and post drought rates. If $-1 < R_{dev} < 0$, the recovery rate (r_{rec}) is slower than the development rate (r_{dev}), indicating that recovery is much slower than the drought development and the system could be vulnerable to a subsequent event. If $R_{dev} < -1$, the recovery is faster than the development, and the system recovers more strongly than it develops to the maximum intensity (Δ_{maxInt}) of the drought event. This interpretation applies to the pre-ratio as

well. For the post-ratio, values between $0 < R_{\text{post}} < 1$ indicate continuous recovery with ongoing post-drought recharge, even though it crossed the specific drought threshold, suggesting that the groundwater is still recovering or a wet season with increased recharge occurs around the same time. Values larger than 1 indicate a sharp recovery that slows after the event, suggesting full recovery and a stronger stabilization of the groundwater levels post-event.

For both pre- and post-event stage ratios, the ratios can be positive or negative. A scenario with a negative post-event ratio could correspond to situations where groundwater levels exhibit high variability, with groundwater levels quickly declining back into another drought event. Similarly, a positive pre-event ratio could correspond to a situation in which the previous drought event was shortly prior, so the drought conditions did not fully alleviate and the system is thus more vulnerable to another dry period. These scenarios are likely occurring in locations with short periods between consecutive drought events or incomplete system recovery before the next event.

R_{dev} and R_{post} are used to create categories that can be linked to the groundwater drought recovery pattern that go beyond describing recovery rates (r_{rec}) in isolation. We followed the ideas on system resilience introduced by Hashimoto *et al* (1982) for water resources systems, where resilience describes how quickly a system recovers from failure or impact. Similar concepts have also been applied for ecosystem studies, regarding ecological resilience (Gunderson 2000). Schwalm *et al* (2017), for example, also implies that longer recovery timescales for ecosystems after drought periods make the system more vulnerable on the short-term. We applied these concepts specifically to the groundwater system by including drought event-specific rates that describe pre-event, development, recovery, and post-event conditions. Based on these insights, we classified locations into resilient (slow development, fast recovery), stable (equal development and recovery), vulnerable (fast development, slow recovery) and unstable (re-entering drought conditions, incomplete recovery) recovery regimes. We define four major categories: resilient, stable, vulnerable and unstable (figure 2). Resilient locations are defined by a strong negative development ratio, indicating a rapid return to normal groundwater levels leading to faster recovery than development. Locations with development ratios within the bounds around -1 (specifically $-(1/0.9)$ and $-(0.9/1)$ used as threshold) are defined as stable, indicating balanced behaviour with equal development and recovery. The threshold was chosen to keep the stable category close to -1 , to avoid overlap with the other classes and their specific drought behaviours (additional exploration of threshold choice can be found in supplementary figure 26). More specifically, the threshold used to describe equal development and recovery will indicate whether drought events and their dynamics are more symmetrical or asymmetrical, and therefore fall into the other categories of the recovery regimes. Vulnerable locations are characterized by a low development ratio, indicating rapid development but slow drought recovery. Unstable locations are defined by a negative post-event ratio, indicating locations that quickly return to drought conditions and do not fully recover on average.

Using this classification, recovery rates (r_{rec}) worldwide can be further assessed, and similar groundwater system behaviours across regions can be identified. Locations with similar behaviour are compared based on potential driving factors, including drought characteristics, geophysical, climatic, and anthropogenic information from the GLOBGM (v1.1). Driving factors across the different regimes are identified by comparing the z-scores of these factors within each regime to the global mean. In terms of drought characteristics, the variables introduced in section 2.2, including drought duration (t_{tot}), drought development (t_{dev}) and recovery time (t_{rec}), maximum intensity (Δ_{maxInt}), number of drought events and the time gap between drought events (event gap, t_{gap}) were taken into account. Furthermore, event-specific rates, such as pre-event (r_{pre}), development (r_{dev}), recovery (r_{rec}), and post-event rates (r_{post}), were considered. For geophysical, climatic and anthropogenic information, we included the variables and parameters available in GLOBGM (v1.1). Parameters on horizontal and vertical conductivity, as well as storage and topography-related aspects, were used as geophysical drivers, while groundwater recharge, mean and standard deviation of groundwater table depth were used to represent climatic drivers and variability. Lastly, groundwater abstractions, which are part of GLOBGM (v1.1), were used to quantify anthropogenic drivers. The list of all available and included variables and parameters can also be found in supplementary figure 27. As the analysis focused solely on groundwater systems, only the variables and parameters listed previously were considered, and they were not necessarily extended to include the full drought propagation from meteorological to hydrological droughts (exceptions are the pre- and post-event windows). For visual comparison and discussion at later stages, we also use the aquifer classification from WHYMAP (Richts *et al* 2011, supplementary figure 5), which is also inherently tied to the model parameterization, as well as the aridity index, indicating water and energy-limited areas, which can also be linked to the groundwater recharge pattern (supplementary figure 6). However, these were not directly considered in the driver analysis and the z-score calculations.

3. Results

3.1. Global average drought recovery pattern

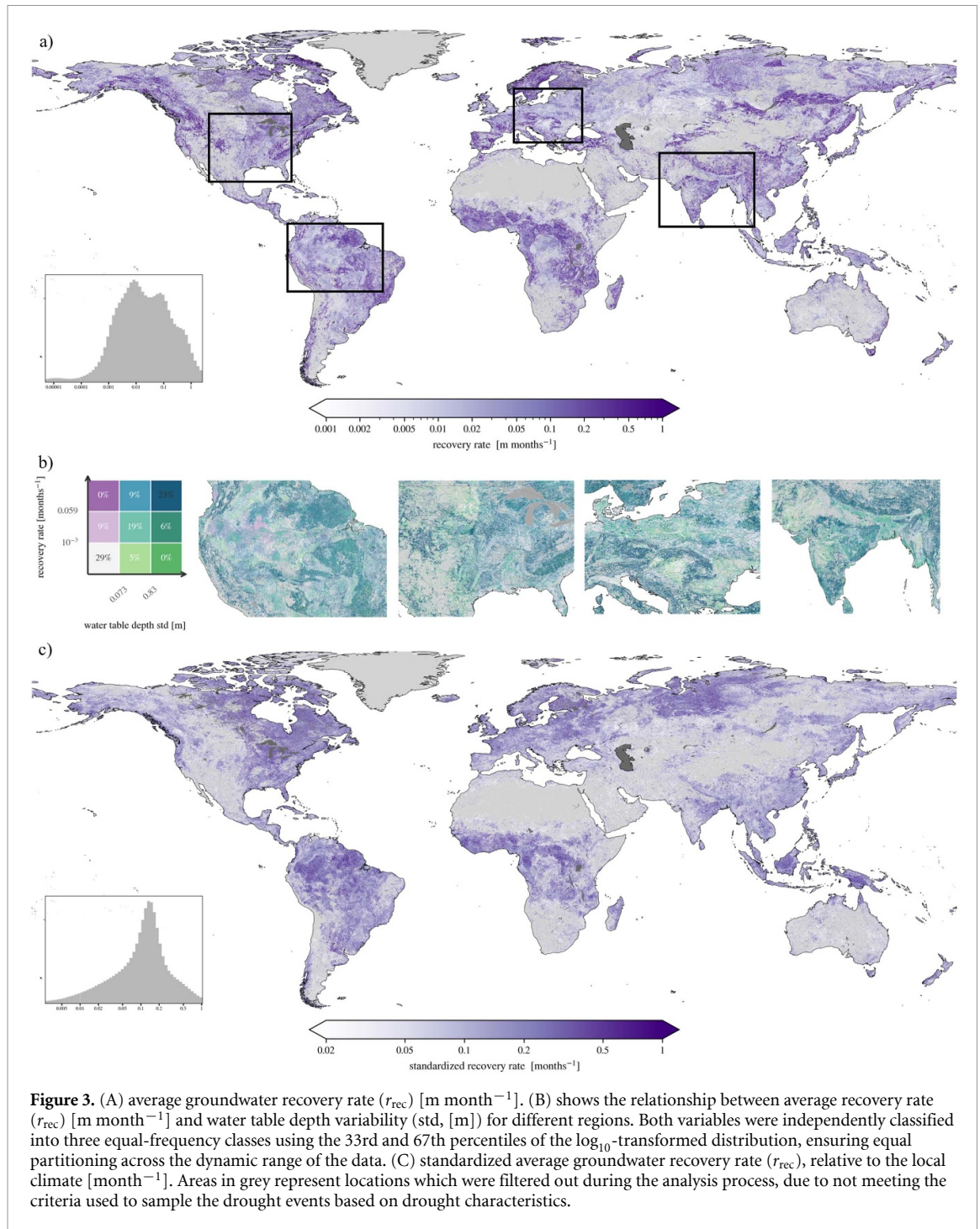
The average spatial pattern of drought recovery around the globe is highly variable, both between and within hydroclimatic regions. Areas in grey represent locations that were filtered out during the analysis process because they did not meet the criteria used to sample drought events based on drought characteristics. These cases often link to locations with strong trends in groundwater table depth, possibly resulting from climate change or anthropogenic influences such as groundwater abstraction, or insufficiently long drought events. Fast recovery rates (r_{rec} , up to 1 m month^{-1}) were observed across all continents to varying degrees, often aligning with mountain ranges but also extending into tropical and equatorial regions (figure 3(a)). Regions with fast recovery, as isolated pockets, showed up within larger spatial regions with mixed recovery rates (r_{rec} , median recovery rate globally $0.018 \text{ m month}^{-1}$). Such heterogeneous patterns were also observed on most continents, for example the Amazon region of South America, throughout Eastern Europe, over large parts of central Africa, and South Asia. On the contrary, Australia shows consistent low to medium recovery rates (r_{rec}) for most of the continent.

Connecting the average recovery rate (r_{rec}) with the standard deviation of water table depth, which was used as a proxy for climate variability, we observe that high water table depth variability often connects to high recovery rates (r_{rec} , figure 3(b)), full map supplementary figure 19). By accounting for the climatological variability by standardizing the average recovery rate (r_{rec}), we allow for a better comparison between regions and identify potential drivers of drought recovery heterogeneity. In figure 3(c), we observe stronger, more coherent recovery patterns, and the large variability within some regions shifted to a main signal. Examples of these shifts were seen in regions that previously showed heterogeneous recovery rates (r_{rec}), such as Eastern Europe and North and South Asia, which now showed relatively fast recovery compared to the median global recovery rate (r_{rec}) (0.12 month^{-1}). Large consistent areas of fast recovery rates (r_{rec}) became visible across the eastern half of North America. Some regions with mixed medium to high recovery rates (r_{rec}), such as the Amazon and central Africa, also showed a more coherent signal with overarching high recovery rates (r_{rec}).

Areas with low standardized average recovery rates (r_{rec}) could visually be linked to climate indicators, such as the aridity index (supplementary figure 6), with areas of slow recovery rates (r_{rec}) often corresponding to arid regions (Australia, Middle East and Central Asia, southern Africa and southern South America, as well as Midwest US) and vice versa (Southeast Asia, Central Africa, Amazon, and Northeast US as well as northern Europe). One exception to this appeared to be India, with a mixed recovery pattern, likely due to the high interannual climate variability and strong anthropogenic signal. The spatial recovery patterns did not necessarily immediately stand out by aquifer types (when considering the WHYMAP aquifer classifications Richts *et al* (2011), supplementary figure 5) with major, complex, and local aquifers) as much as with the aridity index. Exceptions included, for example, bordering complex and local aquifers (India) and major and local aquifers (Central Africa).

The influence of climate and specifically aridity was supported when looking at regions with very fast recovery rates (r_{rec}) in figure 3(c) (e.g. Amazon and Southeast Asia), which could be directly linked to high groundwater recharge rates (supplementary figure 3 and 22). When average recovery rates (r_{rec}) were linked to other drought characteristics, such as average drought duration (t_{tot}), it was found that these two regions often show shorter droughts with high recovery rates (r_{rec} , supplementary figure 23). Furthermore, these events appeared to be more frequent (higher number of drought events and shorter event gaps, t_{gap}) and were relatively intense (average max intensity) compared to other regions (supplementary figure 24 and 25). Complementarily, the development (t_{dev}) and recovery times (t_{rec}) were short, indicating that the groundwater systems are fast responding in general and capable of recovering from a shock to the system back to a neutral groundwater state quickly (Supplementary section 3 with all drought characteristics).

When looking at the global recovery pattern again, these regional findings could be applied to the large scale as well. Fast recovery rates (r_{rec}) were often associated with areas that, on average, have shorter drought durations (t_{tot}). The average development time (t_{dev}) showed a more varied pattern than the recovery rate (r_{rec}), though areas often exhibited fast development and recovery dynamics. The average recovery time (t_{rec}) appeared more homogeneous, with most regions showing similar recovery times (t_{rec}). This suggested that the recovery rate (r_{rec}), which is inherently linked to the maximum intensity (Δ_{maxInt}) of an event, is also strongly linked to conditions earlier in a drought (pre-event and development conditions).



3.2. Drought event dynamic and rates

Continuing the comparison of average recovery rate (r_{rec}) patterns with more event-specific descriptions, such as rates at different event stages (supplementary section 3.2), showed that the development rate (r_{dev}) had a similar spatial pattern as recovery. As previously observed, areas with fast recovery also showed a rapid development rate (r_{dev}), meaning they reached their peak intensity quickly and then recovered rapidly in a symmetrical manner. Similar observations with matching patterns of recovery and development rate (r_{dev}) were made for the pre-event rate (r_{pre}). However, the system approached drought conditions more slowly than the actual development during the drought. Post-event rates (r_{post}) suggested that in most areas around the globe, the system still had the potential to recover more, with slightly positive rates. However, here as well, the rates were lower than those during recovery, indicating more of a stabilisation after the event than an unsuccessful recovery. This, in general, paints a picture where groundwater systems slowly approach drought conditions (pre-event), followed by an increased rate of development of the drought (development rate, r_{dev}) and rapid recovery, which then levels off

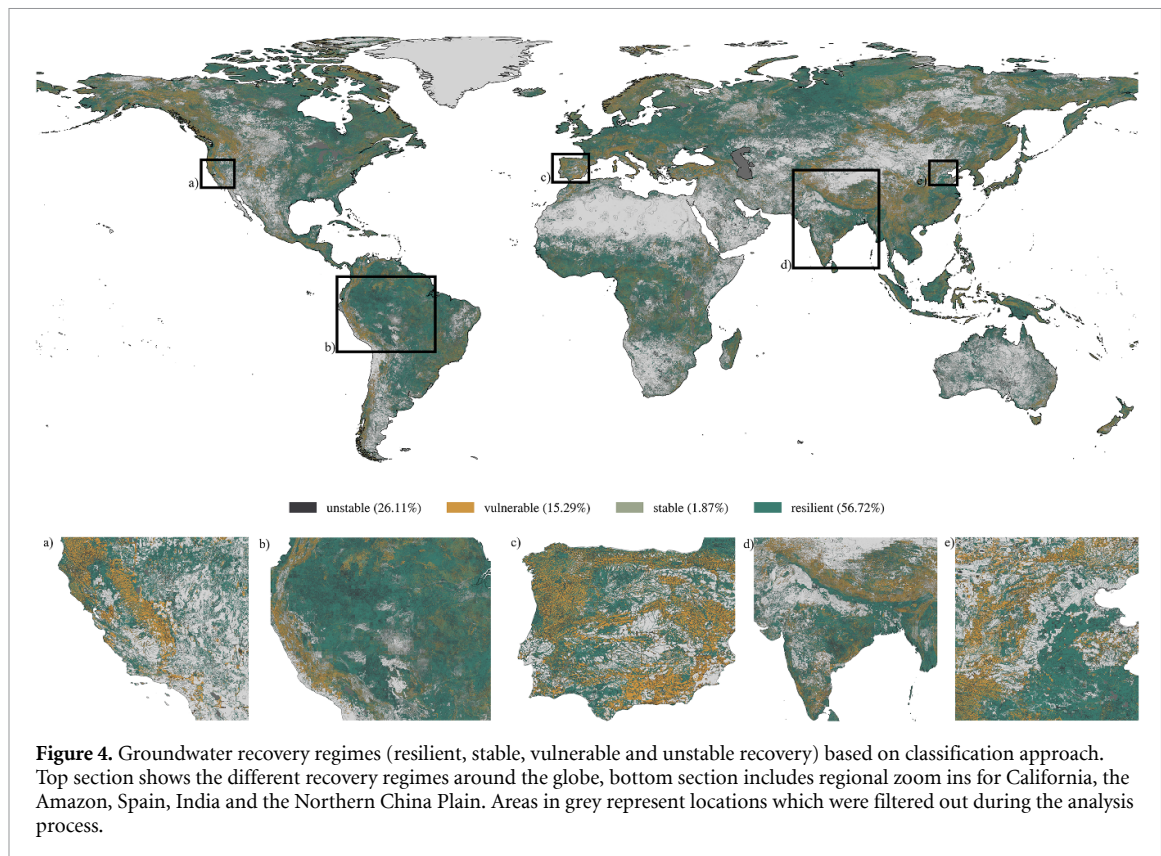


Figure 4. Groundwater recovery regimes (resilient, stable, vulnerable and unstable recovery) based on classification approach. Top section shows the different recovery regimes around the globe, bottom section includes regional zoom ins for California, the Amazon, Spain, India and the Northern China Plain. Areas in grey represent locations which were filtered out during the analysis process.

after the event (post-event rate, r_{post}) is over, making groundwater systems still relatively vulnerable to new drought events. Interestingly, for both pre- and post-event rates (r_{pre} , r_{post}), positive and negative rates were found, indicating locations with unstable behaviour that fell back into drought conditions more quickly, and could also be identified when looking at the event ratios. These specific areas were found around the globe but were sparse and local phenomena (supplementary section 3.2).

3.3. Drought recovery regimes and driving factors

When the event ratios were used and the recovery regimes classification suggested in section 2.3 was applied, it was found that the majority of locations (around 57%) showed resilient behaviour, with faster recovery than development on average (figure 4). Spatially, the resilient areas generally coincided with regions exhibiting high recovery rates (r_{rec}); however, some of these regions, previously identified as high-recovery, were also highlighted in different regime classes. Around 26% of the recovery patterns were linked to areas with unstable recovery behaviour, indicating a quick transition back into drought conditions. Around 15% of locations appeared to show vulnerable recovery patterns, including fast development but slow recovery, and only around 2% exhibited equal development and recovery behaviour, being classified as stable. The latter was characterized by only a few isolated cells, which were spread all over the globe without clear clustering, making it less prominent on the global map. In contrast, the vulnerable regime frequently aligned with distinct topographical features, such as mountain ranges, and might also be linked to specific aquifer properties, for example, karstic regions in Europe.

To identify the driving factors behind these different recovery patterns, we included available geophysical, climatic, and anthropogenic variables from the GLOBGM (v1.1). We then mapped the regime classes to information on conductivity, storage, topographical features, groundwater recharge and water table depth variability, and groundwater abstractions. We computed variable anomalies (z-scores) that quantify deviations from the global mean (supplementary figure 27), which supported these spatial and geophysical observations from figure 4. Additionally, cumulative density functions of a selection of geophysical, climate, and drought characteristics highlighted the differences in regimes (Supplementary section 28). Parameters related to topography, particularly higher elevations relative to the global mean, were prominent in the vulnerable regime and can be identified as characterising features for this regime, along with water table depth (and its variability). Vulnerable and stable regime locations share similarities between different drought characteristics, such as the number of events and frequency, as well as drought development and duration. Compared to the vulnerable regime, the stable regime shows less

strong deviations from the global mean for several geophysical variables. In comparison, the resilient locations seemed to be linked to lower elevations, with lower water table depths and lower water table variability than the global mean. Besides the differences in event rates, storage capacity appeared to be characteristic of this regime, which differs from the latter two, showing slightly more positive anomalies than the stable and vulnerable regime locations.

The unstable regime generally aligns with the global mean across most geophysical parameters. Drought event rates differ from the resilient regime as they show opposite anomalies. In addition, the drought events appear to be shorter, less intense, and more frequent than in all other regimes. Using the variable anomalies to connect the geophysical, climatological and anthropogenic parameters available gave a better understanding of the global spatial pattern we observed in figure 4.

3.4. Regional recovery regime analysis

To further highlight the spatial pattern and drivers between regimes observed globally, representative regional case studies spanning different climates and aquifer types were examined (figures 4(a)–(e)). These regional analyses highlighted the interplay of geophysical, climatic, and anthropogenic factors in shaping regime characteristics and connected local patterns to the broader trends described above.

Across all regions, the vulnerable regime consistently appeared in higher elevation areas with higher horizontal conductivity, lower storage, and greater variability in water table depth (supplementary figures 29 to 33). Drought events in vulnerable and stable regimes tended to have longer event gaps (t_{gap}), slower development (t_{dev}) and recovery times (t_{rec}). With the more regional assessment, it was also possible to see how anthropogenic influences, such as groundwater abstraction, were affecting the regime classification, for example, in arid regions with high human influence, such as Spain and California. In Spain, the vulnerable regime's high elevation, water table depth and its variability were clearly distinguishable. In addition, it was observed that vulnerable and unstable locations share similar groundwater abstraction patterns (supplementary figure 31). California showed a similar behaviour, however more regimes (unstable, stable, and vulnerable) appeared to be affected by groundwater abstractions, and drought development times (t_{dev}) were faster for vulnerable and unstable locations (supplementary figure 29). In mixed arid/wet regions (India and North China Plain), the vulnerable regime maintained its association with previously found geophysical properties. Compared to the other locations, the differences regarding the other parameters were a bit more nuanced (supplementary figure 32 and 33). In wet regions (the Amazon), the vulnerable regime again showed similar behaviour regarding the geophysical properties, while the stable regime was characterized by particularly high storage. Here, the vulnerable regime stood out with elevated water table depth mean and variability. Overall, groundwater abstractions were minimal, and groundwater recharge, compared to other regions, was similar across regimes. Similar to other regions, the vulnerable and stable locations experienced the longest drought intervals and slowest recovery, while the unstable regime was marked by shorter event gaps (t_{gap}). Resilient regimes tend to experience slightly more intense droughts (supplementary figure 30).

In summary, while the vulnerable regime exhibited consistent geophysical and drought characteristics across diverse climatic zones, each region highlighted unique local factors, such as anthropogenic influence or climatic drivers, that further shaped regime dynamics. Overall, these regional observations supported and expanded upon the global regime distinctions, emphasizing the importance of local topography, aquifer properties, climatic variability, and groundwater management in shaping drought regime behaviour and potential attention points in future groundwater drought developments.

4. Discussion

This study assessed the average spatial recovery patterns of groundwater droughts at the global scale by incorporating drought characteristics and drought event rates across the pre-event, development, recovery, and post-event stages. The average event rates were used to classify recovery patterns into four recovery regimes (resilient, stable, vulnerable, and unstable) based on the ratios of drought development (r_{dev}) and recovery rates (r_{rec}). These regimes facilitated the identification of regions with similar groundwater drought dynamics, extending the analysis of drought recovery to also include pre and post-drought conditions, and to provide insights into the underlying geophysical, climatic, and anthropogenic factors that shape the recovery pattern of groundwater droughts.

4.1. Climatic and geophysical drivers of recovery

The average groundwater drought recovery rate (r_{rec}) pattern was strongly influenced by local groundwater variability, as observed when the effect of the year to year groundwater variability was removed

by standardizing the recovery pattern. Locations with high average groundwater drought recovery rates (r_{rec}) were generally associated with areas of high groundwater recharge. The influence of climate, particularly increased precipitation and recharge rates, aligns with previous studies examining groundwater dynamics across various climate zones (Van Loon and Laaha 2015, Singh *et al* 2021, Scanlon *et al* 2023). This is consistent with regional studies demonstrating that wetter climates and higher groundwater recharge fluxes support faster post-drought recovery by replenishing storage deficits more rapidly (Bloomfield and Marchant 2013, Alam *et al* 2021). In contrast, arid and semi-arid regions, where recharge is episodic and strongly tied to intense precipitation events, tend to exhibit slower and more variable recovery (Schreiner-McGraw and Ajami 2021, Scanlon *et al* 2023). While shallow groundwater systems are highly responsive to climatic variability, deeper aquifers often exhibit slower responses to these drivers (Schreiner-McGraw and Ajami 2021, Chen *et al* 2024). However, substantial changes in deeper aquifers are often also linked to human activities such as excessive groundwater pumping (Huang *et al* 2015, Wada 2016, Ren *et al* 2024, Miri *et al* 2026). This interplay between climate-driven recharge and human pressures shapes not only average recovery rates (r_{rec}) but also the broader patterns of recovery behaviour and groundwater vulnerability to droughts across regimes, and motivates the regime-based classification developed here.

4.2. Characteristics of specific recovery regimes

Although the average recovery rate (r_{rec}) pattern aligned with climate-driven groundwater recharge, the introduction of recovery regimes based on the combination of development and post-event conditions, and therefore adding additional drought dynamics information to the analysis, revealed more nuanced regional recovery patterns. Approximately 57% of global regions fell into the resilient recovery regime, characterised by faster drought recovery than development, consistent with areas previously identified as having high recovery rates (r_{rec}). Notably, this category also included regions across various aridity zones and that may not directly be recognized as high groundwater recharge areas. This suggests that aquifer properties such as high storage capacity or connectivity to regional flow systems may buffer against drought development even where recharge is limited, broadly consistent with studies of karst and alluvial systems where structural aquifer properties mediate drought sensitivity (Bloomfield and Marchant 2013, Stoelzle *et al* 2014, MacDonald *et al* 2021), and aligns with the idea that system resilience, as framed by Hashimoto *et al* (1982), depends not only on the magnitude of perturbation but also on the intrinsic capacity of the system to absorb and recover from shocks. Only a small number of locations were classified as stable, representing areas with nearly equal drought development and recovery behaviour. This finding suggests a general asymmetry in drought events, with either longer or shorter development and recovery periods, and could be relaxed if the thresholds in this study were less strict for this category. Furthermore, the stable regime could potentially also be looked at as a more transitional regime if the temporal drought recurrence were assessed in more detail. For example, in locations where drought events might have fallen into a resilient regime in the past and potentially move into a vulnerable regime in the future. Regions that exhibited faster drought development than recovery were classified as having a vulnerable recovery regime and accounted for approximately 15% of the study area. These locations were distinguished by geophysical properties and water-table depth variability, as indicated by the recovery regime map, the higher variability in anomalies and different geophysical, climatic, and drought characteristics that were shown to affect this regime. Regional analysis showed that conductivities, storages, and topographic variables were often consistent across regions and regimes, while other variables, including water table depth variability, groundwater recharge, and abstractions, remain location-specific and are likely to alter drought characteristics locally. In particular, aquifer transmissivity and storage coefficients govern drought memory effects and the degree to which past deficits persist and delay recovery, meaning that even modest shifts in recharge or abstraction rates, could be sufficient to push vulnerable locations across regime boundaries under future conditions (Hellwig *et al* 2021, Chen *et al* 2024). de Graaf *et al* (2024) also showed that groundwater abstractions can have a significant impact on groundwater surface-water interactions, especially in alluvial aquifers, leading to increased river capture, putting surface water systems under more pressure as well. The combination of these different properties was also highlighted by regional studies documenting prolonged recovery timescales following the Millennium Drought in Australia and multi-year droughts in Mediterranean Europe, where low-permeability lithologies and deep water tables were shown to extend recovery considerably beyond the duration of the forcing event itself (Yang *et al* 2017, Hellwig *et al* 2020, Schreiner-McGraw and Ajami 2021). These aspects are therefore interesting to further investigate, as these might be crucial to understand when and how locations classified as vulnerable recovery might switch into other recovery regimes under future conditions. Beyond vulnerable locations, a further subset of regions showed an even more critical characteristic with a tendency to go back into drought conditions before

full recovery could occur. A substantial number of locations were classified within the unstable recovery regime, indicating a tendency to revert to drought conditions. These locations were distributed globally and were not exclusively associated with areas of significant groundwater abstraction, despite the recognized importance of abstraction in groundwater drought development (Wendt *et al* 2020, Scanlon *et al* 2023). Many of the locations affected by very strong trends in both water table depth and groundwater abstractions have been filtered out in the initial stages of the drought analysis based on their drought characteristics, and also correspond to the locations highlighted previously as regions known to represent locations with water table trends (van Jaarsveld *et al* 2026). Therefore, locations where drought vulnerability could be dominated by groundwater abstractions leading to a strong groundwater trend might be underrepresented in the current analysis. In contrast to the resilient regime, unstable locations were less spatially coherent and more spread out. Some of these areas may have experienced frequent drought events with short intervals between events, which can influence pre- and post-event conditions and potential groundwater memory lags. This would be consistent with compound and clustered drought events, where successive events with insufficient recovery time between drought intervals progressively erode groundwater storage. These drought clusters are likely to increase under climate change (Van Loon and Laaha 2015, Akl *et al* 2025). The global and spatially dispersed distribution of unstable locations, spanning both heavily abstracted basins and regions with naturally limited groundwater recharge, suggests that this regime arises from multiple interacting drivers rather than a single dominant cause, and underscores the importance of considering pre- and post-event conditions alongside drought intensity and duration when assessing groundwater system resilience.

4.3. Methodology and modelling uncertainties

The importance of pre- and post-event conditions also relates to the methodology used to calculate average drought characteristics and to the definitions and implementation of drought events. Previous studies, such as those by (Parry *et al* 2016a), examined drought recovery in streamflow time series and defined drought events using fixed or variable thresholds, often merging events separated by a minimum time gap (Parry *et al* 2018, Margariti *et al* 2019). This approach is fitting for streamflow, which exhibits greater temporal fluctuations than groundwater due to its faster response to increased precipitation. In this study, the focus on monthly groundwater simulations led to the analysis of individual events without merging short gaps. Testing in selected areas indicated that short event gaps (t_{gap}) frequently occurred in regions with numerous brief droughts. These brief drought events of two months duration were excluded from the analysis, as these were too short to be considered. However, remaining drought events that might have occurred within the six-month period included in the pre- and post-event variables might still have affected the pre- and post-event conditions, which ultimately link to the unstable category. For these events, having a focus on the temporal, event-based recovery rate (r_{rec}) pattern would be needed to get a better understanding of the pre- and post-event conditions, which are important aspects for drought development and recovery and likely even more in the future with increased drought frequency due to climate change and human influences and the potential of groundwater systems reaching a potential point of not recovering. These methodological considerations also interact with uncertainties inherent in the global groundwater simulations underpinning the analysis.

Besides uncertainties coming from methodological considerations, additional uncertainties can be associated with global groundwater simulations (Reinecke *et al* 2025). Although physically-based models improve process understanding and incorporate human influences, and how these differ from satellite-based products such as GRACE (Li *et al* 2019), large-scale modelling efforts have demonstrated both underestimation and overestimation of groundwater in the past compared to observations (Döll *et al* 2016). Recent advances in hyper-resolution modelling, such as GLOBGM (v1.1) by van Jaarsveld *et al* (2026), included improved calibration and validation efforts, and downscaling procedures, and while uncertainties remain, their performance has improved, and validation efforts show skilful predictions for shallow and deep aquifers. Additionally, physically-based models allow for extrapolation to regions where local observations are absent. This makes groundwater models better suited to assess a wide range of spatial scales compared to coarse-resolution satellite products, as their application for small aquifers and highly seasonal water storage changes can be challenging (Longuevergne *et al* 2010, Famiglietti *et al* 2011, Scanlon *et al* 2012, Wada 2016). Examples, where previous global groundwater models were used include for example groundwater-dependent ecosystems (Otoo *et al* 2025), groundwater drought response at regional scale (Hellwig *et al* 2020), as well as assessing groundwater pumping limits (de Graaf *et al* 2019, 2024, Bierkens *et al* 2021). Despite inherent limitations, such models provide a base for exploring large-scale processes and trends, which can be relevant for transboundary aquifers and groundwater resource challenges, and can complement other groundwater research approaches. Furthermore, such models can

enable us to study future groundwater changes by using different climate and socio-economic forcings (van Jaarsveld *et al* 2026).

4.4. Summary and outlook

With groundwater resources under increasing pressure from both climatic and anthropogenic forces, understanding the effects of climate extremes, such as droughts, is important for assessing how these important freshwater resources recover from these events and whether they will cross tipping points in the future. The recovery regime framework introduced here extends beyond previous large-scale groundwater drought studies, which have largely focused on drought frequency, duration, and severity or on long-term depletion trends (Li *et al* 2019, Scanlon *et al* 2023, Jasechko *et al* 2024), by explicitly characterizing the groundwater drought dynamics, with a focus on the average recovery pattern and identifying regions where this recovery might show groundwater systems at structural risk. Compared to catchment- and regional-scale studies that have linked recovery timescales to specific hydrogeological controls, including aquifer permeability in the UK (Bloomfield and Marchant 2013), soil moisture–groundwater coupling in central Europe (Stoelzle *et al* 2014), groundwater recharge lag times in semi-arid environments (Schreiner-McGraw and Ajami 2021), and combined climate and abstraction effects in Asia and North America (Alam *et al* 2021, Chen *et al* 2024), the present study provides a global-scale high-resolution synthesis of these behaviours and situates them within a coherent classification framework applicable across diverse hydroclimatic and anthropogenic settings. In this study, we assessed the spatial patterns of groundwater drought recovery and introduced the use of recovery regime classification, describing how robust or vulnerable groundwater drought recovery is across different locations and which factors may drive it. Assessing the temporal, event-based characteristics of groundwater drought recovery could help to better understand the importance of the different event stages and conditions. Examining specific rates at different stages of drought events could provide deeper insights, particularly regarding the influence of pre- and post-drought conditions and the role of human activities such as groundwater abstraction (Van Loon and Laaha 2015). The role of abstraction could be of particular interest for future assessments with the ongoing trend of groundwater use and groundwater depletion. These trends do provide additional challenges when it comes to quantifying drought events in locations with strong trends (Wanders *et al* 2015). Our current analysis filtered out these locations with strong long-term depletion trends via the drought selection process to focus on drought event dynamics. However, groundwater abstractions at sub-trend timescales, for example, intensified pumping during drought periods to compensate for reduced surface water availability, could substantially extend recovery timescales and may partly explain the occurrence of the unstable regime in irrigated agricultural regions (Wada 2016, MacAllister 2024, Ren *et al* 2024). Ultimately, increased groundwater abstractions can also have strong impacts on groundwater surface-water interactions and therefore surface water availability by increased river capture (de Graaf *et al* 2024). This feedback, whereby drought conditions trigger increased groundwater use, which in turn prolongs or deepens the deficit, represents a growing risk as climate change and population growth simultaneously increase demand and the frequency of meteorological drought (Scanlon *et al* 2023, Akl *et al* 2025). However, the biggest challenge for including this dynamic abstraction information in this work is the coarse spatial and temporal resolution of global abstraction data. Previous studies at the local and regional scale have shown that abstractions can significantly alter drought characteristics, while improved water management strategies, including managed aquifer recharge, have proven effective in mitigating drought impacts and increasing groundwater storage (Wendt *et al* 2021, Jasechko *et al* 2024). Identifying areas with vulnerable groundwater drought recovery in this study may inform the mapping of areas with increased drought stress, complementing regional assessments, and could profit from such water management interventions if not already implemented. Given that vulnerable and unstable regimes frequently cross national boundaries, these findings also carry implications for transboundary groundwater governance, where coordinated monitoring and management strategies will be essential to prevent unilateral exploitation from pushing shared aquifer systems past their resilience and sustainability limits.

Under climate change projections, many regions are expected to experience increased drought frequency and intensity, reduced recharge in already water-stressed areas, and shifts in precipitation seasonality (Scanlon *et al* 2023). These changes will likely not affect all recovery regimes equally, resilient regions may maintain their buffering capacity if recharge remains sufficient, whereas vulnerable and unstable regions could face increased risk of transitioning toward states of persistent drought or irreversible storage depletion, reaching a sustainability tipping points. Identifying and monitoring regions at risk of such regime transitions could therefore be a priority for groundwater management policies and adaptive water management frameworks. Additionally, large-scale physically-based models that incorporate human influences could be used to simulate future scenarios under increasing anthropogenic and

climatic pressures, evaluating whether regions may transition between recovery regimes or lose recovery capacity altogether. Coupling the recovery regime classification developed here with scenario-based projections from global groundwater models driven by CMIP6 climate forcing could provide an additional tool for mapping how the global distribution of resilient, vulnerable, and unstable groundwater systems may evolve over the coming decades. Such analyses could further evaluate the mitigation potential of interventions, including managed aquifer recharge, demand reduction, and land-use change, helping to prioritise actions in regions where recovery capacity is most at risk and where the window for effective management is narrowest.

5. Conclusions

This study identified and classified global patterns in groundwater drought recovery using hyper-resolution simulations from the global groundwater model GLOBGM (v1.1) model for the period 1960–2019. Our results show that spatial variability in groundwater recovery rates was predominantly governed by climate variability, though local geophysical factors also played a significant role. Classifying locations into different recovery regimes (resilient, stable, vulnerable, and unstable) based on average event-specific rates showed that areas with similar average recovery rates can differ in their recovery regimes and underlying drivers. Vulnerable regime locations were linked to geophysical features, often related to topography, and high climate variability. Both vulnerable and stable regimes tend to have fewer but longer drought events compared to resilient and unstable regime locations, while the latter was characterised by many droughts in rapid succession, suggesting a tendency to return to drought conditions rapidly after the previous event has ended. Resilient regime locations were characterised by a more stable climate. By linking these regimes to drought characteristics and environmental parameters, we provided new insights into the drivers of drought resilience. These results underscore the importance of considering both climatic and geophysical factors in groundwater management and can help guide efforts to strengthen resilience by identifying groundwater systems at risk of crossing critical thresholds under future drought and abstraction pressures.

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Data availability statement

GLOBGMv1.1 simulations based on van Jaarsveld *et al* (2026) can be accessed via YODA, Utrecht University's research data management service. For this study, the historical reference simulations have been used. The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.24416/UU01-AKSHOX>.

Supplementary data 1 available at <https://doi.org/10.1088/3033-4942/aea63d/data1>.

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